Smith's class-level census, counted by isomorphism class

research/smith_iso_granularity_20261015 · 2026-10-15 · protocol PROTOCOL.md frozen at 05f3d9203 before any data

1. Question

The 2026-10-14 census (research/smith_prop24_native_20261014) found that a tractable L-class usually holds a model on which Smith's construction fully succeeds, but that the class-level share C lies **below** an independence null that scattered successes over census **models**. The open item in problems/smith-kernel-isogeny-class-walk.md asked whether the deficit is a structural invariant of the class or the sparsity of small classes. This study tests the simplest sparsity explanation: full_success is a property of a curve up to $\mathbb{F}_p$-isomorphism, and one curve accounts for many census models, so the model-level null over-counts the independent coins per class.

2. Verdicts (frozen wording)

prime	observed C	iso-level null: median [2.5%, 97.5%] (999 perms)	stratified by orbit size too (secondary)	verdict
$p = 7$	0.4304	0.4579 [0.4545, 0.4611]; $P(\text{null} \le C) = 0$	0.4582 [0.455, 0.4611]	**residual deficit beyond granularity**
$p = 11$	0.475	0.4864 [0.4858, 0.4871]; $P(\text{null} \le C) = 0$	0.4861 [0.4855, 0.4867]	**residual deficit beyond granularity**
overall				**residual deficit beyond granularity**

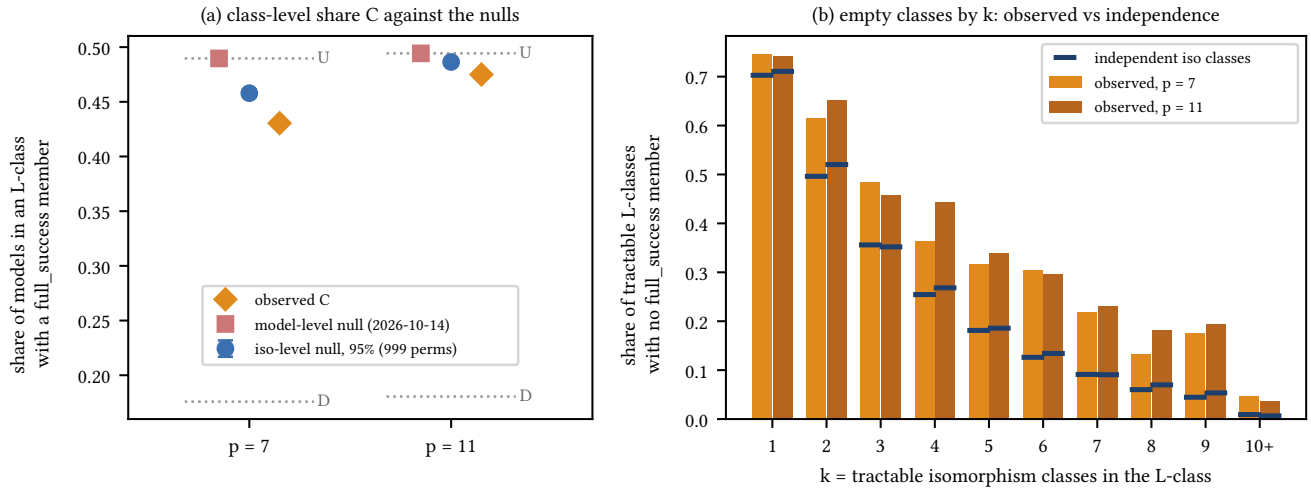


Figure 1: (a) Observed C against the model-level null of 2026-10-14 (square; its 95% range is a single value) and the iso-level null of this study (circle, 95% range); dotted lines are the direct share D and the ceiling U. (b) Share of tractable L-classes with no full_success isomorphism class, by the number k of tractable isomorphism classes they hold; bars observed, ticks the expectation under independent isomorphism classes with the per-stratum success rate. Source figures/plot.py, data data/summary_p*.json.

3. Measured

quantity (exact)	p = 7	p = 11
census models / isomorphism classes / L-classes	1411788 / 34227 / 5231	35431220 / 324562 / 21701
tractable isomorphism classes / with full_success	16823 / 5927	160010 / 58053
median tractable isomorphism classes per tractable L-class	6	14
D / C / U (reproduce 2026-10-14)	0.1758 / 0.4304 / 0.4897	0.1805 / 0.475 / 0.4943
tractable L-classes with no full_success: observed / expected (independent iso classes)	560 / 394.8	1210 / 720.3

4. Findings

- **Measured.** full_success is an isomorphism invariant on every census model (control C2), and the census models fall into 34227 isomorphism classes at $p = 7$ and 324562 at $p = 11$; a tractable L-class holds a median of 6 and 14 tractable isomorphism classes.
- **Measured.** With the isomorphism class as the unit, the null median of C drops from U to 0.4579 ($p = 7$) and 0.4864 ($p = 11$). **Derived**, $\frac{U - \text{null median}}{U - C}$: counting curves instead of models accounts for 54% and 41% of the gap between U and the observed C.
- **Measured, frozen rule.** The observed C is below every one of the 999 permutations at both primes, also when permutations are stratified by orbit size: **residual deficit beyond granularity** at $p = 7$ and $p = 11$. Tractable L-classes with no full_success member: 560 observed against 395 expected ($p = 7$), 1210 against 720 ($p = 11$). The excess appears at every k (Figure 1b), so it is not confined to the smallest classes.
- **Post hoc, exploratory (not registered).** Within each #S(H) stratum the isomorphism-level success rate barely moves with the 2-adic valuation of $\#J(\mathbb{F}_p) = L(1)$: at $p = 11$ it stays within 0.239–0.271 for $\#S(H) = 1$ and 0.549–0.584 for $\#S(H) = 3$ (post_hoc_by_stratum_and_v2_of_J in the summaries). That is far too small to double the number of empty classes, so the 2-power part of the group order is not the hidden class invariant.
- **Interpretation (labelled).** Successes cluster within L-classes: the isomorphism classes of one L-class are positively correlated in whether Smith's two square conditions pass. Something finer than χ_π mod 2 and coarser than the curve is shared within a class; a candidate not tested here is the Galois structure of $J[4]$ or of the 2-adic Tate module, which the L-polynomial does not fix. For the walk

lead this means that the reachable set is C, not the granularity-corrected null, and that the classes with no `full_success` member are a real feature of the data rather than an artefact of how models were counted.

5. Validation (controls of the protocol; all must pass)

control	**result**
C1: per-L-class model, tractable and `full_success` counts equal `census_p*_classes.csv`	$p = 7$: 0 of 5231 rows differ; $p = 11$: 0 of 21701 differ.
C2: `full_success`, stratum and L-key constant on every isomorphism class (every model checked)	inconsistent models: 0 ($p = 7$), 0 ($p = 11$); orbit images outside the squarefree census: 0, 0.
C3: orbits partition the census	unvisited models: 0, 0.
instrument	**valid**

Orbits are enumerated exactly: every non-Weierstrass point of $\mathbb{P}^1(\mathbb{F}_p)$ is sent to infinity, the leading coefficient is normalised by a square (its class decides the twist), the octic is depressed and $x \to ux$ is applied for every $u \in \mathbb{F}_p^*$; this covers all of $\mathrm{PGL}_2(\mathbb{F}_p)$ modulo translations. C2 is the check that `full_success` is a curve invariant on every one of the census models, not only on sampled ones.

6. Scope

Exact enumeration at $p = 7, 11$, degree-8 models only. Nothing here builds, prices or bounds an isogeny walk, or says anything at cryptographic sizes or in characteristic 2. A rejected granularity hypothesis means the deficit is not explained by counting curves instead of models; it does not identify the structural cause.

7. Reproduce

```
cargo build --release --example smith_iso_granularity
B=target/release/examples/smith_iso_granularity; D=research/smith_iso_granularity_20261015/data
$B --p 7 --out $D; $B --p 11 --out $D        # 17 s and 15 min on 2 vCPU
$B summarize --dir $D 7 11
python research/smith_iso_granularity_20261015/figures/plot.py   # layout only
```

Source: B. Smith, J. Cryptology 22 (2009), arXiv:0806.2995. Prior: `research/smith_prop24_native_20261014/`.